Foundations of Biomedical Informatics
Quiz - 3

Date - 08 April 2026 **Time - 90 minutes**
Max. Marks - 100 **Name:**

SECTION - A

Multiple Choice Questions (MCQs)
15*4=60

1. Which combination is most essential for a CDSS to generate recommendations?
   A. Patient data + Medical knowledge + Algorithms
   B. Only medical images + Cloud storage
   C. Patient data + Billing information
   D. Only doctor's notes
2. Which of the following is a key challenge in building CDSS?
   A. Increasing hospital room size
   B. Converting medical expertise into computable models
   C. Replacing all paper records with printers
   D. Reducing the number of clinicians

3. CDSS is increasingly integrated with which of the following systems?
   A. Social media platforms
   B. Electronic Health Records and Hospital Information Systems
   C. Weather forecasting systems
   D. Banking databases

4. What is the main lesson from **Anscombe's Quartet**?
   A. Different datasets always have different summary statistics
   B. Similar summary statistics can still represent very different data patterns
   C. Correlation always proves causality
   D. Linear relationships are always stronger than nonlinear ones

5. Why is **confounding** important in causal analysis?
   A. It increases the sample size
   B. It creates a hidden influence that may falsely suggest a causal relationship
   C. It removes all noise from data
   D. It converts correlation into causation automatically

6. What is the main purpose of **Bayes' Rule** in biomedical reasoning?
   A. To remove all uncertainty from clinical data
   B. To update probability estimates when new evidence is observed
   C. To calculate only linear correlations
   D. To replace data visualization.

7. If the conditional probability P(Marker | Cancer) is very high, which of the following MUST also be true?
   A. The probability P (Cancer | Marker) is also very high.
   B. The marker is a highly specific indicator of cancer.
   C. The test has high sensitivity, it rarely misses the disease in sick patients.
   D. The marker is a "False Positive" for healthy patients.

8. Which of the following best defines **conditional probability**?
   A. Probability of two independent events occurring together
   B. Probability of an event occurring given that another event has already occurred
   C. Probability that always equals 1
   D. Probability of a false positive only

Assertion-Reasoning

A. Both Assertion and Reason are true, and Reason correctly explains Assertion
B. Both true, but Reason does NOT explain Assertion
C. Assertion true, Reason false
D. Assertion false, Reason true

9. **Assertion (A):** A Clinical Decision Support System (CDSS) may generate accurate recommendations, yet clinicians may still choose not to follow them.
   Reason (R): Effective CDSS implementation depends not only on algorithms, but also on usability, trust, and integration with clinical workflow. **A**

10. **Assertion (A):** A test for lung tar with 100% Sensitivity is the perfect tool for a doctor to confirm that a patient definitely has lung cancer.
    Reason (R): Sensitivity only measures the ability to catch all sick people; it does not measure the ability to exclude healthy people who might also have high tar. **D**

11. **Assertion (A):** Probabilistic decision systems are useful in biomedical applications because many clinical decisions must be made even when data is incomplete or uncertain.
    Reason (R): These systems estimate the likelihood of outcomes using probability rather than relying only on fixed deterministic rules. **A**

12. Confounding variables can create misleading associations between two variables in biomedical data. **True**
13. In probabilistic decision systems, a probability of 1 always means the prediction is uncertain. **False**
14. The language standard commonly used for sharing clinical data between hospitals is **_HL7_**.
15. Data organized with each row as one observation is called **_long_** format.

SECTION-B

Long Answer Questions **4*10 = 40**

1. Mention the main goals of CDSS in healthcare (2). Name any two key challenges in building CDSS (2). Draw a flowchart showing the working of the CDSS. (6)

Answer:

Main Goals: (any 2)

1. **Improve diagnostic accuracy:** CDSS assists clinicians by analyzing patient data (symptoms, history, lab results) and comparing it with vast medical knowledge bases. It helps in identifying possible diseases, reducing diagnostic errors, and supporting early detection of conditions.
2. **Support treatment selection:** CDSS provides evidence-based recommendations for treatment plans, including drug selection, dosage, and therapy options. It ensures that clinicians follow standardized clinical guidelines and best practices.
3. **Improve patient safety:** CDSS helps prevent medical errors by generating alerts such as drug–drug interactions, allergies, and incorrect dosages. This reduces adverse events and improves overall quality of care.

The **key challenges** are: **(any 2)**

1. **Knowledge encoding:**

   Medical knowledge from literature and guidelines must be converted into structured, machine-readable formats (rules, ontologies, or models). This is complex because medical knowledge is vast, constantly evolving, and often ambiguous.

2. **Uncertainty/incomplete clinical data:**

   Patient data may be missing, noisy, or uncertain. CDSS must still provide reliable recommendations despite these limitations, which is a major technical challenge.

3. **Workflow integration :**

   CDSS must seamlessly integrate into existing hospital systems (like EHRs) without disrupting clinical workflows. Poor integration can lead to low adoption.

4. **Usability/trust by clinicians:**

   If the system is difficult to use or provides unclear recommendations, clinicians may ignore it. Building trust requires transparency, accuracy, and ease of use.

```
        Medical Literature
                |
Clinical Guidelines ──────┐
                |
Patient Data (EHR) ───▶ Clinician ──▶ Decision
                |
Hospital Protocols ──────┘
      ↑
    CDSS
(Synthesizes information)
```

2. A longitudinal study observes that patients with high accumulations of Lung Tar (Biomarker X) also exhibit the most advanced stages of Lung Cancer (Disease Y). a. Does the presence of high lung tar alone prove it is the biological cause of lung cancer? Why or why not? (3). b.Describe how the "Directionality Problem" complicates this study. Could the cancer itself be causing the tar buildup? (3) .c.In this specific study, what would constitute a False Positive result? (2). d.What could be the study compared to claiming causality? (2)

   Answer:
   a. **No.** Correlation is not causation. Tar and cancer may both be independent effects of a shared cause (smoking).
   b. **Y → X:** It can be possible only if cancer develops first and restricts bronchial pathways or impairs the lung's cleaning mechanism, the cancer might be the reason tar is stuck there, not the other way around.
   c. A patient who has high lung tar but **no** cancerous cells present
   d. “Smokers with tar” vs “non-smoker’s with tar” would be the best comparison to claim causality.

3. A rare disease has a prevalence of 1%. A diagnostic test has high sensitivity but moderate specificity.
   (a) Explain how Bayes' Theorem updates disease probability after a positive test. **(4)**
   Bayes’ Theorem updates the probability of disease after a positive test by combining the prior probability (prevalence = 1%) with test characteristics (sensitivity and specificity).
   A positive test increases the probability of disease, but since the disease is rare, the post-test probability may still remain low.

   ✔ Key points to award marks:
   - Prior probability = prevalence (1%)
   - Uses sensitivity and specificity
   - Probability increases after positive test
   - Still may be low due to rarity

(b) Why can a positive result still have low clinical reliability? **(3)**
Even with a positive test, reliability can be low because the disease is rare. Most people are healthy, and with **moderate specificity**, many **false positives** occur. This leads to a **low positive predictive value (PPV)**.

✔ Key points to award marks:
- Low prevalence → most are healthy
- False positives due to moderate specificity
- Low PPV mentioned

(c) What does this imply for screening programs? **(3)**
Screening programs must use **confirmatory tests** after a positive result. They are more effective in **high-risk populations**. False positives can cause **anxiety, unnecessary treatment, and increased cost**.

✔ Key points to award marks:
- Confirmatory testing needed
- Target high-risk / higher prevalence groups
- Harms of false positives (cost/anxiety/overtreatment)

Part	Question Component	Expected Answer Elements	Marks
(a)	Bayes' Theorem explanation	Defines Bayes' concept (post-test depends on prior + test accuracy)	1
		Identifies prevalence (1%) as prior probability	1
		Explains role of sensitivity and specificity	1
		Interprets that probability increases but may remain low due to rarity	1
		Subtotal	**4**
(b)	Clinical reliability of positive result	Explains low prevalence → most individuals are healthy	1
		Moderate specificity → false positives	1
		Mentions low PPV (positive predictive value)	1
		Subtotal	**3**
(c)	Implications for screening programs	Need for confirmatory testing	1

		Screening better in high-risk / high-prevalence groups	1
		Mentions harms (anxiety, cost, overdiagnosis)	1
		Subtotal	**3**
		TOTAL	**10**

4. A hospital collects daily blood pressure readings for patients over a month. The data can be stored in long format or wide format.
   (a) Explain long format and wide format in your own words. **(2)**
   (b) Draw a simple example of long format and wide format for patient "P1" with 3 days of blood pressure readings. **(2)**
   (c) List 2 advantages of long format and 2 disadvantages. **(2)**
   (d) List 2 advantages of wide format and 2 disadvantages. **(2)**
   (e) Explain how the choice of format affects statistical analysis and machine learning. **(1)**
   (f) Give one scenario where long format is preferred and one scenario where wide format is preferred. **(1)**

Ans.(a) Long format vs Wide format
Long format:
In long format, data is arranged such that each row corresponds to a single observation at a specific time point. This means that if a patient has multiple readings over time, their ID will appear in multiple rows. Time or measurement instance is stored as a separate column.
Wide format:
In wide format, each row represents a single patient, and repeated measurements (such as daily blood pressure readings) are stored in separate columns. Each column corresponds to a different time point.

(b) Example for patient P1 (3 days BP readings)
Long format:
Patient_ID Day Blood_Pressure
P1 1 120
P1 2 125
P1 3 118
Here, patient P1 appears in multiple rows, one for each day.

Wide format:
Patient_ID BP_Day1 BP_Day2 BP_Day3
P1 120 125 118
Here, patient P1 appears only once, and all measurements are stored in columns.

(c) Advantages and disadvantages of Long format

Advantages:
Suitable for longitudinal and time-series analysis:
It naturally represents repeated measurements over time, making it ideal for studying trends and temporal changes.
Compatible with advanced statistical models:
Many statistical techniques such as mixed-effects models, survival analysis, and repeated measures ANOVA require data in long format.
Disadvantages:
Larger dataset size:
Since each observation is stored in a separate row, the dataset can become very large and harder to manage.
Less intuitive for quick viewing:
It is difficult to directly compare all measurements of a single patient without filtering or grouping.

(d) Advantages and disadvantages of Wide format
Advantages:
Easy to interpret and compare:
All measurements for a patient are visible in a single row, making it easier for quick inspection and reporting.
Suitable for basic machine learning models:
Many ML algorithms expect fixed input features, which aligns well with wide format.
Disadvantages:
Inflexible with varying time points:
If patients have different numbers of observations or missing days, the structure becomes inconsistent.
Not suitable for many statistical analyses:
Advanced statistical models often require reshaping the data into long format before analysis.

(e) Effect on statistical analysis and machine learning
The choice of format directly impacts analysis:
Long format allows models to account for within-patient variation over time, making it essential for longitudinal studies.
Wide format treats each measurement as an independent feature, which is useful for standard machine learning models but may ignore temporal relationships.

(f) When to use which format
Long format preferred:
When analyzing ICU patient vitals over time or studying disease progression, where time dependency is important.
Wide format preferred:
When building a predictive model using fixed inputs (e.g., BP readings from Day 1–3 to predict outcome).